

Dhairya Kamalia

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Education

Arizona State University, MS in Computer Software Engineering	Aug 2024 – May 2026
Gujarat Technological University, BE in Computer Engineering	Aug 2020 – May 2024

Experience

Software Engineering Intern, BSL, New Jersey, United States	May 2025 – Aug 2025
- Developed an Object Detection Module (ODM) for architectural floor plans, optimizing a YOLO-based model to enable automated, efficient detection of building elements within an architectural pipeline - Engineered a multimodal RAG pipeline by integrating GPT-4 Vision and custom embeddings to match detected objects with product data, demonstrating scalable system design. - Engineered a computer vision pipeline utilizing Hough Circle Transform and contour analysis alongside bilateral filtering and Otsu's thresholding to achieve robust and reliable detection of door tags. - Applied advanced preprocessing (thresholding, morphological operations, noise removal) to maximize recognition accuracy and runtime efficiency of the parsing module.	
Software Developer Intern, Atyantik Technologies PVT LTD, India	April 2023 – April 2024
- Engineered a high-accuracy 99.47percent image classification model using a CNN, GANs, and Vision Transformers; designed the system for low-latency inference in catalog matching - Developed a custom text-to-text LLM using BERT, RoBERTa, and LLaMA 2, integrated with GPT-4 Vision for multi-modal AI analysis. - Built AI-driven mood board generation and matching algorithms leveraging LLMs, focusing on computational efficiency and data optimization.	
Software Engineer Intern, KrishaWeb IT and Services, Colorado, United States	Oct 2022 – Nov 2022
- Developed a distributed Gym Management System using Java, Spring Boot microservices, and MySQL, applying object-oriented programming (OOP) principles and implementing JWT-based authentication to ensure secure and scalable operations. - Engineered resilient relational database schemas and integrated Stripe payment workflows, prioritizing fault tolerance, transactional integrity, suited for high-availability, large-scale systems.	

Projects

Multimodal Visual Question Answering (VQA) Model	Transformers and NLP
- Designed a 197M parameter multimodal AI model, architecting the system to optimize data flow between text and image processing units for VQA. - Implemented fusion methodologies using text encoders (BERT) and image encoders (ViT) to inc AI-based Q/A.	
Distributed Analytics Platform	RAG and LLM
- Built RESTful microservices for real-time data processing and incorporated CloudWatch and ELB for robust monitoring and load balancing. - Utilized S3 for data storage and RDS (PostgreSQL) for management, ensuring system reliability and scalability	

Technologies

Courses: Machine Learning, Data Structures and Algorithms, Operating Systems, Cloud Computing
Tools: PyTorch, TensorFlow, Keras, OpenCV, LangChain, Generative AI, NLP, Vision Transformers
Programming: Python, Java, C#, C, C++, JavaScript, Golang,